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BioSim Talk #5

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21st August 2026 (Friday)

3.30 – 5.00 pm

Institute for Protein Research
University of Osaka (Suita Campus)
4th floor seminar room

Correlating Nucleic Acid Structures with Water and Environmental Factors: From Ion Binding to Collective Water Dynamics

Understanding how nucleic acid structures interact with their surrounding water environment is essential for explaining their stability in crowded cellular conditions. Across three complementary studies, we examined this problem from the perspectives of ion binding, hydration thermodynamics, and collective solvent dynamics. Molecular dynamics simulations revealed that choline ions bind to DNA in a base-pair-composition-dependent manner, stabilizing A–T base pairs while destabilizing G–C pairs through distinct groove interactions. In parallel, Grid Inhomogeneous Solvation Theory (GIST) demonstrated that crowding agents such as ethylene glycol destabilize the hydration shell around hairpin duplexes, whereas G-quadruplexes maintain a thick, robust hydration layer that resists cosolute intrusion. Finally, distance-resolved dipole correlation analysis showed that water reorientation is a collective process involving approximately seventy molecules across multiple coordination shells, providing a microscopic origin for the broad Debye relaxation observed in the gigahertz range. Together, these studies present an integrated picture in which DNA stability emerges from the interplay between specific ion recognition, hydration-shell thermodynamics, and long-range cooperative water dynamics. This framework highlights water not as a passive background but as an active architectural component that shapes nucleic acid structures under biologically relevant conditions.

Link for online participation via Zoom:

Meeting ID: 822 7529 0770

Passcode: 164283

Please inform us if you will be participating online or joining our Slack channel

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